

ALEENA A L

Electronics and Communication Engineer

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PROFILE SUMMARY

B.Tech Electronics and Communication Engineering graduate with hands-on experience in embedded systems, UAV systems, IoT, and robotics. Skilled in embedded programming, PCB design, flight controller integration, and wireless communication protocols, with a strong foundation in C, Embedded C, Python, and hardware-software co-design. Experienced in developing reliable, high-performance embedded systems through end-to-end hardware-software integration.

SKILLS

- Programming languages: C, Python, Embedded C.
- Embedded Systems: ESP32, Arduino, Xiao, STM32, 8051
- IoT: ThingSpeak, Blynk, Thingsboard, NodeMCU
- PCB Design, Circuit Design: Altium, LTspice, EasyEDA, Proteus, KiCad.
- UAV: Drones, RC planes, Betaflight, ArduPilot, iNav
- RTOS, interrupt handling
- IDEs: Arduino IDE, Keil, MATLAB, Simulink.
- CAD/Simulation: SolidWorks, ANSYS, XFLR5, Simulink, Onshape
- Protocols: SPI, I2C, UART, MQTT, LoRa
- Robotics: ROS2 basics, sensor integration

Soft skills: Effective communication, Critical thinking, Leadership, Teamwork, Time management, Problem solving, Documentation.

EDUCATION

- **B.Tech in Electronics and Communication Engineering** (GPA: 8.22/10) 2022 - 2026
Mar Athanasius College of Engineering, Kothamangalam
- **Senior Secondary** (Percentage: 97.75%) 2020-2022
Govt Boys HSS, Tripunithura

ACHIEVEMENTS

NPTel — Drone Systems and Control | Elite Certificate | Score: 95% | Top 2% in India.
AIR 4 — ASME IAM3D Drone Competition
AIR 8 — ASME Student Design Competition
2nd Prize — Electronics Project Exhibition, MACE

PROJECTS

- **Quadcopter Design & Development:** Designed and fabricated a custom quadcopter with a 3D-printed airframe using SolidWorks and ANSYS; configured and tuned flight controller using iNav, integrating GPS and optical flow sensors for stable position-hold flight.
- **IoT-Based Communication System:** communication system with mesh network integration using NodeMCU for remote areas, with limited or no network connectivity.
- **Wearable Safety Device for Women** (IEEE Funded Project): Built a wearable safety device using a Seeed XIAO controller, integrating GPS tracking, accelerometer-based emergency detection, automated SMS alerts, and vital-sign monitoring.
- **Water Tank Cleaning Robot:** Developed autonomous water tank cleaning robot using Raspberry Pi and Pi-Cam with ML-based dirt detection; integrated web interface for remote monitoring and manual override control.
- **High-precision Infusion Syringe Pump with microlitre flow control:** Achieved $\pm 10 \mu\text{l}$ precision by developing a high-precision infusion syringe pump with microlitre-level flow regulation using embedded control and a ball-screw-driven actuation mechanism.
- Additional Projects: RFID car parking system (8051) | 8x8x8 LED Cube (custom PCB) | RC plane.

EXPERIENCE

- ASME IAM3D Drone Making Competition (AIR 4) – Led team as Vice-Captain.
- Operational Team Member, IoT CLUB ExeCom (June 2023 – May 2024)
- GIRLATHON Hackathon – IoT-Based Communication System (Top 15 out of 250 Teams).
- MathWorks Minidrone Competition 2025 – Created Simulink Models and control systems for autonomous drone navigation.
- SAE DDC – RC Plane Design: Performed aerodynamic analysis (XFLR5) and structural simulation (ANSYS) for RC plane.
- Internship at Keltron (2025)
Designed and developed an IoT-enabled intrusion detection system using ESP8266, PIR sensors, and camera modules, incorporating motion-triggered image acquisition, event processing, and real-time alert generation. Integrated MQTT protocol and ThingSpeak cloud services to facilitate low-latency data communication, real-time monitoring dashboards, remote system management, and secure storage of sensor data and security events.